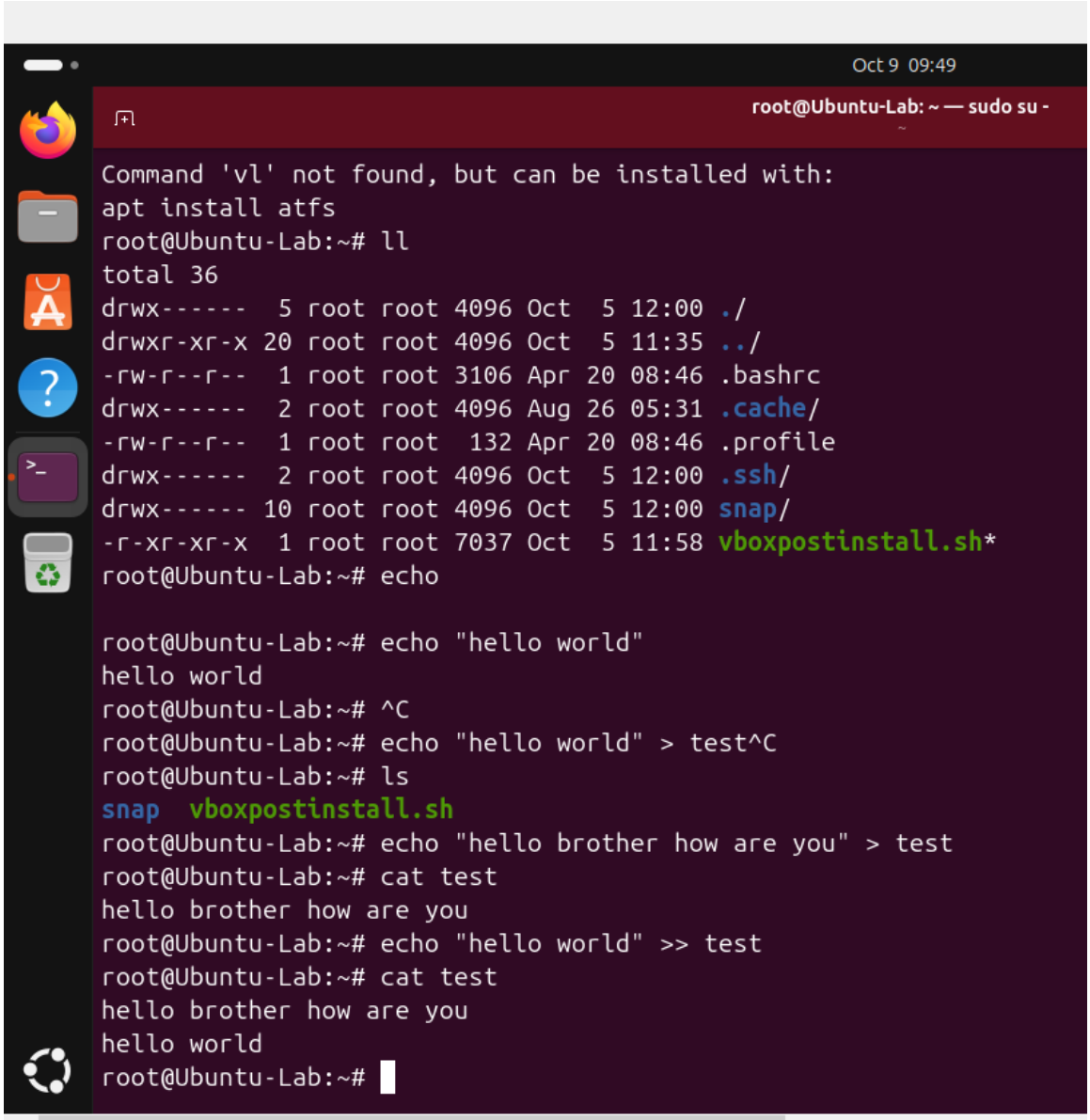
Practice 2 Linux/Ubuntu

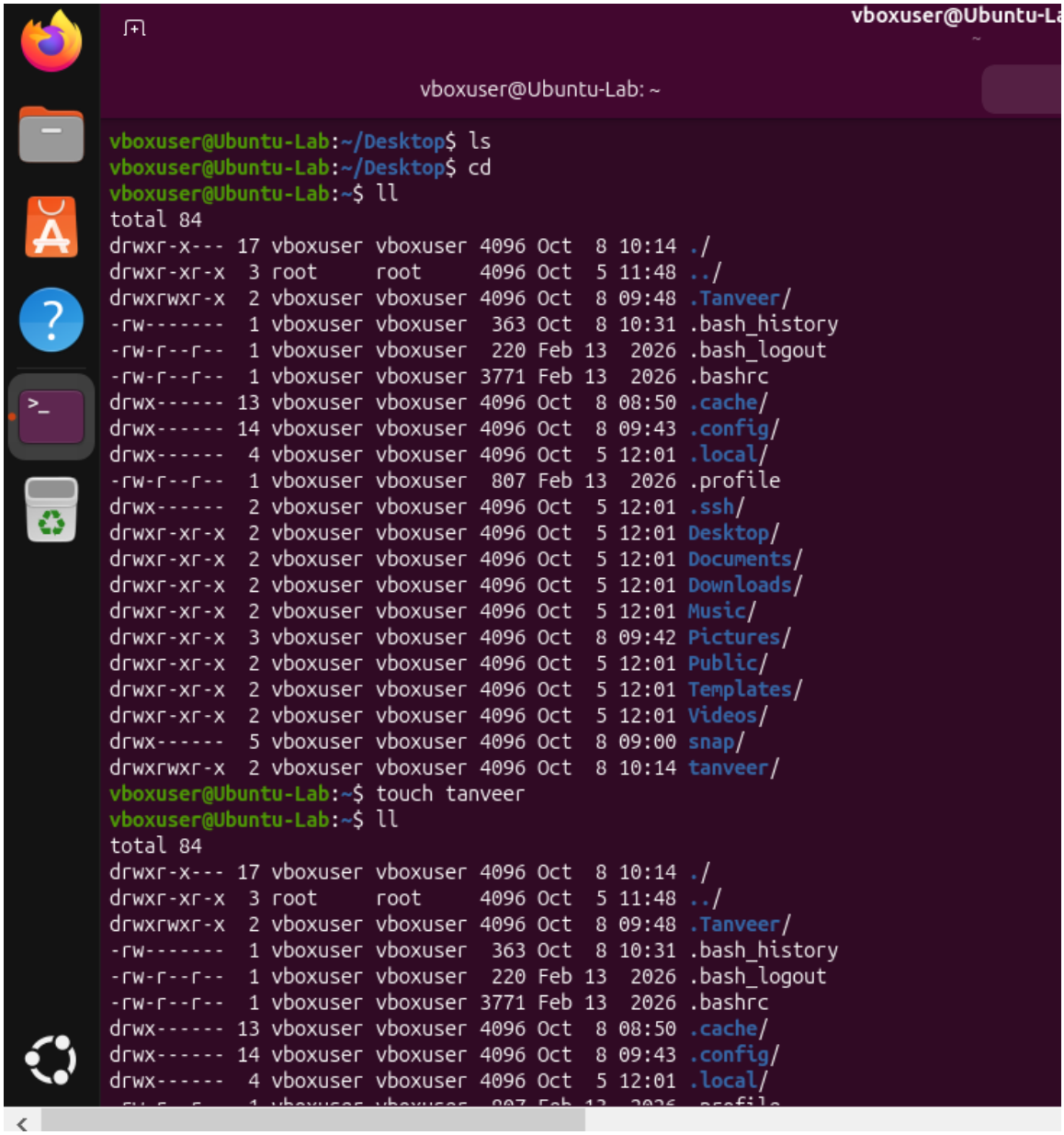
1: Exploring Directory Contents & File Creation (root user)

- ls -la: Lists all files (including hidden ones starting with a dot) in the current directory along with permissions, owners, sizes, and timestamps. You viewed files in the root user's home directory.
- echo: Prints arguments to the standard output. Without text, it just prints a blank line.
- echo "hello world": Prints the text 'hello world' to the terminal screen.
- echo "hello world" > test: Redirects the output text into a file named 'test' (creating it or overwriting it). Interrupted with ^C before finishing cleanly.
- ls: Lists the files in the current working directory, confirming snap and vboxpostinstall.sh exist.
- echo "hello brother how are you" > test: Writes the string into a file named 'test', completely overwriting existing content.
- cat test: Concatenates and displays the contents of the file 'test' (Printed 'hello brother how are you').
- echo "hello world" >> test: Appends (adds to the end) the string to the existing file 'test' without overwriting it.
- cat test: Displays updated contents of 'test' showing both lines.
- vl alphabet: Attempted to run a non-existent command, returning a command not found error.



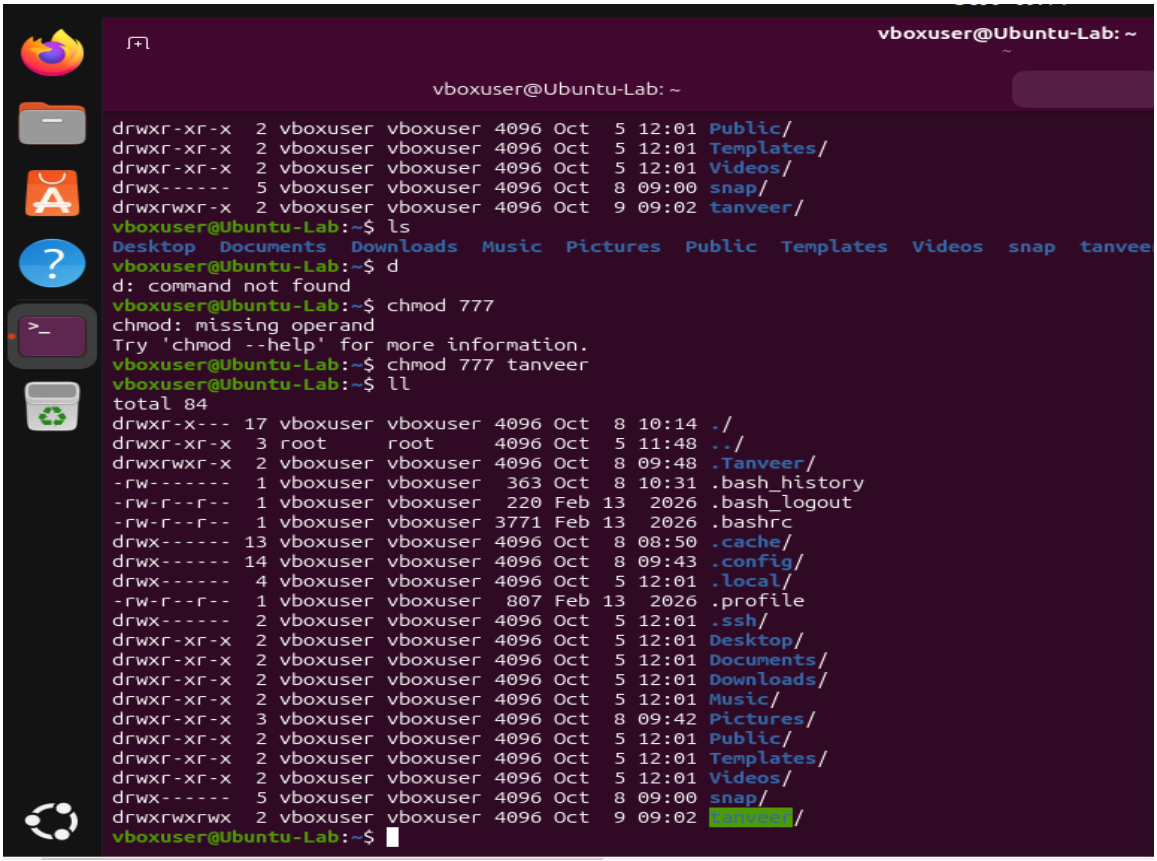
2: User Switch & Basic Navigation (vboxuser)

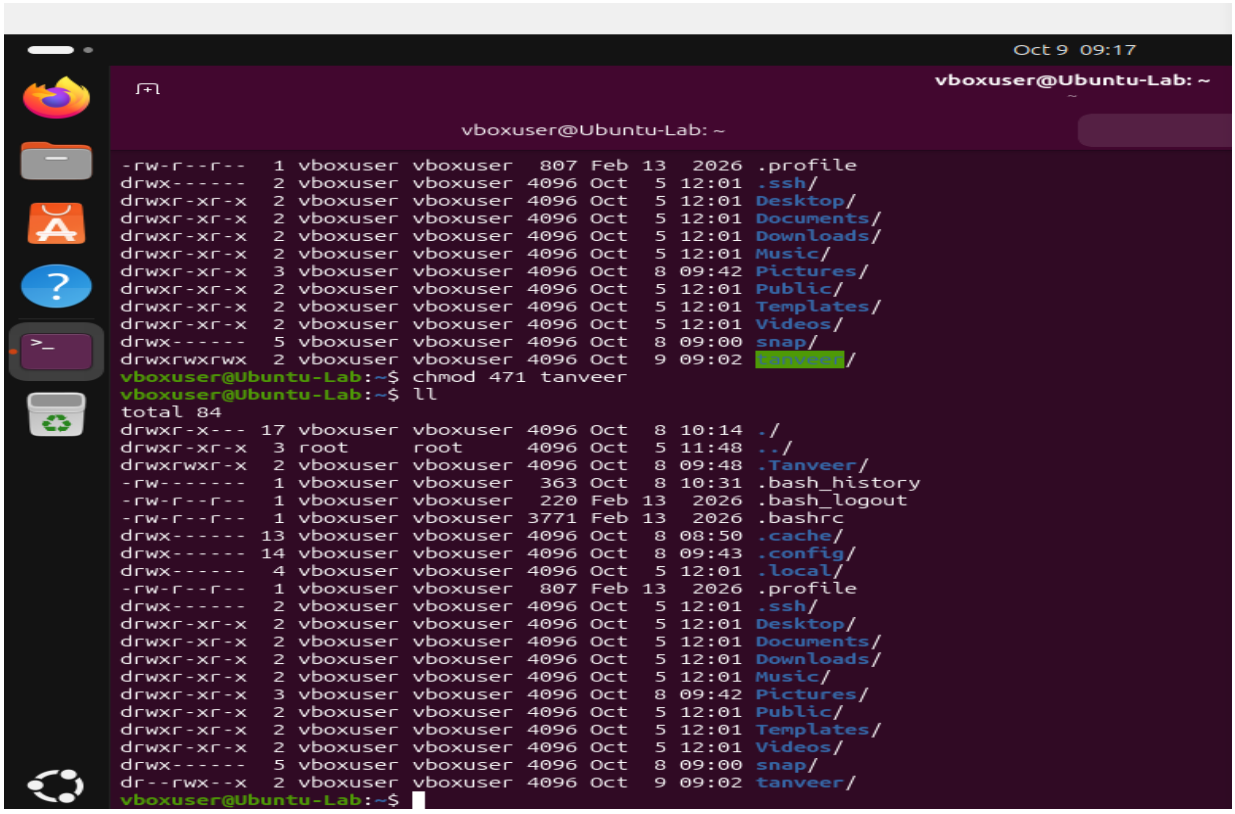
- touch tanveer: Creates an empty file named 'tanveer' or updates its timestamp if it already exists.
- ll: A common shorthand alias for ls -l (long listing format). Displayed detailed file permissions.
- ls: Lists visible files and folders in your home directory.
- d: Entered a shorthand or mistyped command, returning a 'command not found' error.
- cd: Change directory (with no arguments, returns you to your home directory).



3: Changing File Permissions (chmod)

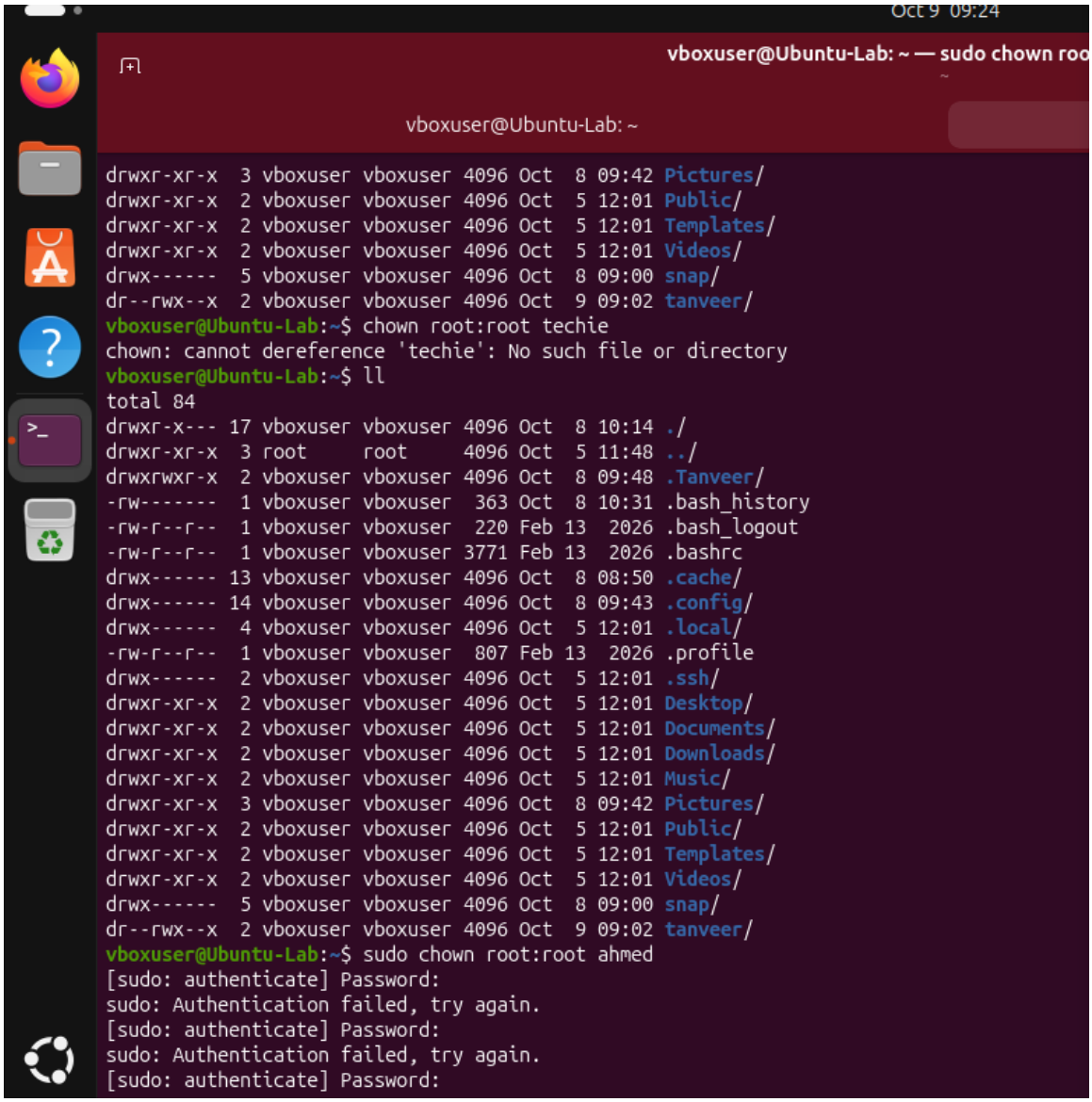
- chmod 777: Changes file access permissions, but failed because no target file was specified.
- chmod 777 tanveer: Grants full read, write, and execute permissions to everyone for the file 'tanveer'.
- chmod 471 tanveer: Sets specific numeric permissions (Owner: read-only, Group: execute-only, Others: execute-only).
- ll -ltrh: Lists files in long format, reverse time order, sorted by most recent, with human-readable file sizes.
- ls -ll: Redundant flags for listing directory contents in long format.

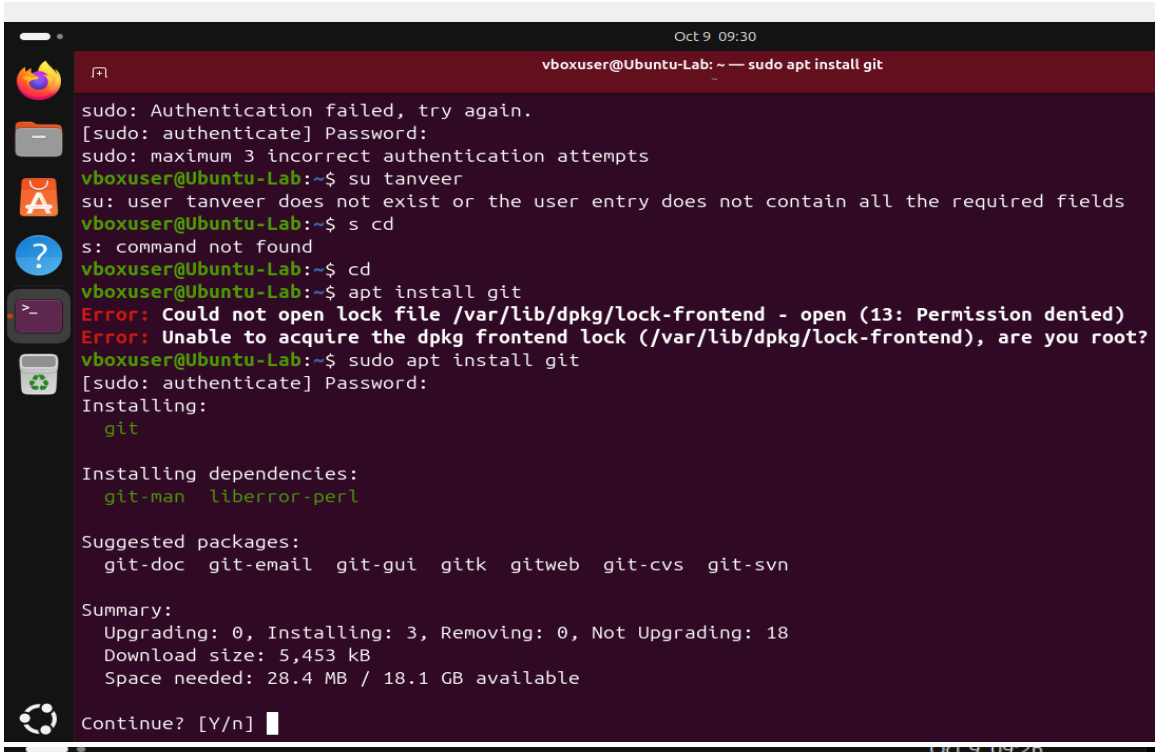
- 
vboxuser@Ubuntu-Lab: ~
vboxuser@Ubuntu-Lab: ~
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Public/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Templates/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Videos/
drwx------ 5 vboxuser vboxuser 4096 Oct 8 09:00 snap/
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct 9 09:02 tanveer/
vboxuser@Ubuntu-Lab:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap tanvee
vboxuser@Ubuntu-Lab:~$ d
d: command not found
vboxuser@Ubuntu-Lab:~$ chmod 777
chmod: missing operand
Try 'chmod --help' for more information.
vboxuser@Ubuntu-Lab:~$ chmod 777 tanveer
vboxuser@Ubuntu-Lab:~$ ll
total 84
drwxr-x--- 17 vboxuser vboxuser 4096 Oct 8 10:14 ./
drwxr-xr-x 3 root   root   4096 Oct 5 11:48 ../
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct 8 09:48 .Tanveer/
-rw------- 1 vboxuser vboxuser 363 Oct 8 10:31 .bash_history
-rw-r--r-- 1 vboxuser vboxuser 220 Feb 13 2026 .bash_logout
-rw-r--r-- 1 vboxuser vboxuser 3771 Feb 13 2026 .bashrc
drwx------ 13 vboxuser vboxuser 4096 Oct 8 08:50 .cache/
drwx------ 14 vboxuser vboxuser 4096 Oct 8 09:43 .config/
drwx------ 4 vboxuser vboxuser 4096 Oct 5 12:01 .local/
-rw-r--r-- 1 vboxuser vboxuser 807 Feb 13 2026 .profile
drwx------ 2 vboxuser vboxuser 4096 Oct 5 12:01 .ssh/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Desktop/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Documents/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Downloads/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Music/
drwxr-xr-x 3 vboxuser vboxuser 4096 Oct 8 09:42 Pictures/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Public/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Templates/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Videos/
drwx------ 5 vboxuser vboxuser 4096 Oct 8 09:00 snap/
drwxrwxrwx 2 vboxuser vboxuser 4096 Oct 9 09:02 tanveer/
vboxuser@Ubuntu-Lab:~$

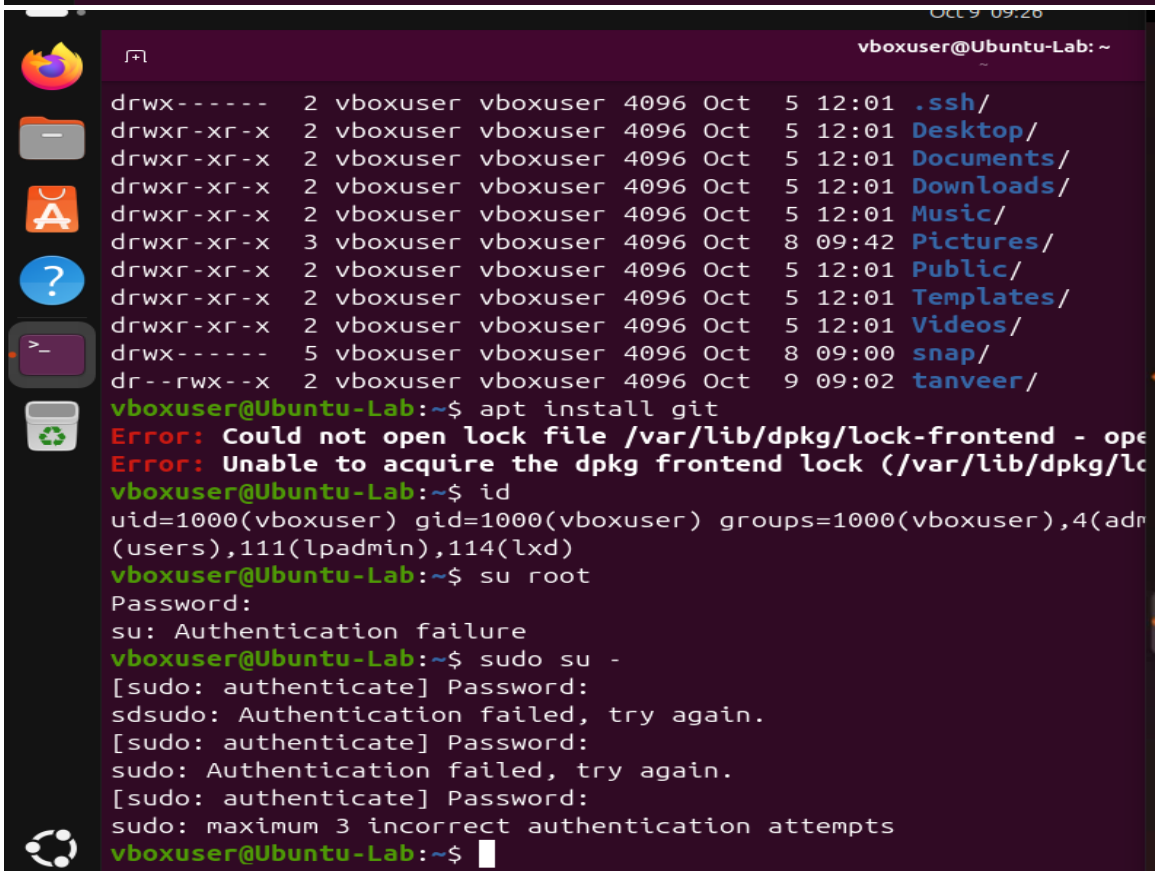

Oct 9 09:17
vboxuser@Ubuntu-Lab: ~
vboxuser@Ubuntu-Lab: ~
-rw-r--r-- 1 vboxuser vboxuser 807 Feb 13 2026 .profile
drwx------ 2 vboxuser vboxuser 4096 Oct 5 12:01 .ssh/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Desktop/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Documents/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Downloads/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Music/
drwxr-xr-x 3 vboxuser vboxuser 4096 Oct 8 09:42 Pictures/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Public/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Templates/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Videos/
drwx------ 5 vboxuser vboxuser 4096 Oct 8 09:00 snap/
drwxrwxrwx 2 vboxuser vboxuser 4096 Oct 9 09:02 tanveer/
vboxuser@Ubuntu-Lab:~$ chmod 471 tanveer
vboxuser@Ubuntu-Lab:~$ ll
total 84
drwxr-x--- 17 vboxuser vboxuser 4096 Oct 8 10:14 ./
drwxr-xr-x 3 root   root   4096 Oct 5 11:48 ../
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct 8 09:48 .Tanveer/
-rw------- 1 vboxuser vboxuser 363 Oct 8 10:31 .bash_history
-rw-r--r-- 1 vboxuser vboxuser 220 Feb 13 2026 .bash_logout
-rw-r--r-- 1 vboxuser vboxuser 3771 Feb 13 2026 .bashrc
drwx------ 13 vboxuser vboxuser 4096 Oct 8 08:50 .cache/
drwx------ 14 vboxuser vboxuser 4096 Oct 8 09:43 .config/
drwx------ 4 vboxuser vboxuser 4096 Oct 5 12:01 .local/
-rw-r--r-- 1 vboxuser vboxuser 807 Feb 13 2026 .profile
drwx------ 2 vboxuser vboxuser 4096 Oct 5 12:01 .ssh/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Desktop/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Documents/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Downloads/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Music/
drwxr-xr-x 3 vboxuser vboxuser 4096 Oct 8 09:42 Pictures/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Public/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Templates/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Videos/
drwx------ 5 vboxuser vboxuser 4096 Oct 8 09:00 snap/
dr--rwx--x 2 vboxuser vboxuser 4096 Oct 9 09:02 tanveer/
vboxuser@Ubuntu-Lab:~$

4: System Files & Privilege Escalation Attempts

- cat /etc/passwd: Displays the list of all local user accounts and system daemons registered on the Ubuntu system.
- chown / sudo chown: Attempts to change file ownership to root; failed because the target files did not exist or sudo authentication failed.
- apt install git: Attempts to install Git; failed initially with a Permission Denied / Lock File Error because standard users lack root privileges.
- id: Prints user ID and group ID information for the current user (vboxuser, UID 1000).
- su root / sudo su -: Commands intended to switch user accounts or log in as superuser; failed due to authentication errors.
- sudo apt install git: Runs package installation with administrator privileges via sudo. Successfully prompted for password and installed git.




Oct 9 09:30
vboxuser@Ubuntu-Lab: ~ — sudo apt install git
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
vboxuser@Ubuntu-Lab:~$ su tanveer
su: user tanveer does not exist or the user entry does not contain all the required fields
vboxuser@Ubuntu-Lab:~$ s cd
s: command not found
vboxuser@Ubuntu-Lab:~$ cd
vboxuser@Ubuntu-Lab:~$ apt install git
Error: Could not open lock file /var/lib/dpkg/lock-frontend - open (13: Permission denied)
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontend), are you root?
vboxuser@Ubuntu-Lab:~$ sudo apt install git
[sudo: authenticate] Password:
Installing:
git
Installing dependencies:
git-man liberror-perl
Suggested packages:
git-doc git-email git-gui gitk gitweb git-cvs git-svn
Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 18
Download size: 5,453 kB
Space needed: 28.4 MB / 18.1 GB available
Continue? [Y/n]


vboxuser@Ubuntu-Lab: ~
drwx------ 2 vboxuser vboxuser 4096 Oct 5 12:01 .ssh/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Desktop/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Documents/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Downloads/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Music/
drwxr-xr-x 3 vboxuser vboxuser 4096 Oct 8 09:42 Pictures/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Public/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Templates/
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct 5 12:01 Videos/
drwx------ 5 vboxuser vboxuser 4096 Oct 8 09:00 snap/
dr--rwx--x 2 vboxuser vboxuser 4096 Oct 9 09:02 tanveer/
vboxuser@Ubuntu-Lab:~$ apt install git
Error: Could not open lock file /var/lib/dpkg/lock-frontend - ope
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lc
vboxuser@Ubuntu-Lab:~$ id
uid=1000(vboxuser) gid=1000(vboxuser) groups=1000(vboxuser),4(adr
(users),111(lpadmin),114(lxd)
vboxuser@Ubuntu-Lab:~$ su root
Password:
su: Authentication failure
vboxuser@Ubuntu-Lab:~$ sudo su -
[sudo: authenticate] Password:
sdsudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
vboxuser@Ubuntu-Lab:~$